

5.5 For each of the following two time-series regression models, and assuming MR1-MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}$$

- c. Provide an intuitive explanation for these results.

a.

$$\begin{aligned} \sum_{t=1}^T (t - \bar{t})^2 &= \sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T} \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{\left[\frac{T(T+1)}{2}\right]^2}{T} \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4} \\ &= \frac{T(T+1)}{2} \left[\frac{2T+1}{3} - \frac{T+1}{2} \right] \\ &= \frac{T(T+1)}{2} \left[\frac{4T+2-3T-3}{6} \right] \\ &= \frac{T(T+1)}{2} \left[\frac{T-1}{6} \right] \\ &= \frac{T(T+1)(T-1)}{12} = \frac{T(T^2-1)}{12} \end{aligned}$$

$$\begin{aligned} \text{Var}(b_2|\mathbf{x}) &= \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2} \\ &= \frac{\sigma^2}{\frac{T(T^2-1)}{12}} = \frac{12\sigma^2}{T(T^2-1)} \end{aligned}$$

$$\therefore \lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T^3 - T} = 0$$

b. $x_t = 0.5^t$

$$\sum_{t=1}^T (x_t - \bar{x})^2 = \sum_{t=1}^T x_t^2 - \frac{(\sum_{t=1}^T x_t)^2}{T}$$

$$\begin{aligned} \textcircled{1} \sum_{t=1}^T x_t^2 &= \sum_{t=1}^T (0.5^t)^2 = \sum_{t=1}^T 0.25^t \\ &= \frac{0.25(1 - 0.25^T)}{1 - 0.25} = \frac{0.25}{0.75}(1 - 0.25^T) \\ &= \frac{1}{3}(1 - 0.25^T) \end{aligned}$$

$$\begin{aligned} \textcircled{2} \sum_{t=1}^T x_t &= \sum_{t=1}^T 0.5^t = \frac{0.5(1 - 0.5^T)}{1 - 0.5} \\ &= \frac{0.5}{0.5}(1 - 0.5^T) = 1 - 0.5^T \\ \Rightarrow \sum_{t=1}^T (x_t - \bar{x})^2 &= \frac{1}{3}(1 - 0.25^T) - \frac{(1 - 0.5^T)^2}{T} \end{aligned}$$

$$\text{Var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\frac{1}{3}(1 - 0.25^T) - \frac{(1 - 0.5^T)^2}{T}}$$

$$T \rightarrow \infty, 0.25^T \rightarrow 0, 0.5^T \rightarrow 0, \frac{1}{T} \rightarrow 0$$

$$\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\frac{1}{3}} = 3\sigma^2 \neq 0.$$

- c. $x_t = t$ represents a linear time trend. As T grows, the values keep expanding, causing the total variation of x to approach infinity. This means every new observation brings substantial new information, effectively driving the estimation variance to 0. Hence, it's consistent.

$x_t = 0.5^t$ decays toward 0 extremely rapidly. After the first observation, subsequent values of x_t are practically zero. Therefore, adding more data points ($T \rightarrow \infty$) adds virtually no new variation or information about how x affects y . Since the total variation in x hits an asymptote (it caps at $\frac{1}{3}$), the uncertainty in our estimate never completely vanishes (remaining stuck at $3\sigma^2$). Hence, it's inconsistent.

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$K=3 \quad N-K=60 \quad \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \end{matrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- $\beta_2 = 0$
- $\beta_1 + 2\beta_2 = 5$
- $\beta_1 - \beta_2 + \beta_3 = 4$

a. $t\text{-statistic} = \frac{b_2 - \beta_2}{\text{se}(b_2)} = \frac{3}{2} = 1.5$

$$t_{60, 0.975} = 2$$

$$\therefore 1.5 < 2$$

$\therefore$ Do not reject H_0 .



c.

$$\begin{aligned} t\text{-statistic} &= \frac{(b_1 - b_2 + b_3) - 4}{\text{se}(b_1 - b_2 + b_3)} \\ &= \frac{(2 - 3 - 1) - 4}{\sqrt{16}} = \frac{-6}{4} = -1.5 \end{aligned}$$

b.

$$t\text{-statistic} = \frac{b_1 + 2b_2 - 5}{\text{se}(b_1 + 2b_2)} = \frac{3}{\sqrt{11}} \approx 0.9045$$

$$\begin{aligned} \text{Var}(b_1 + 2b_2) &= \text{Var}(b_1) + 4\text{Var}(b_2) + 4\text{cov}(b_1, b_2) \\ &= 3 + 4(4) + 4(-2) \\ &= 11 \end{aligned}$$

$$\therefore 0.9045 < 2 \quad \therefore \text{Do not reject } H_0$$

$$\begin{aligned} \text{Var}(b_1 - b_2 + b_3) &= \text{Var}(b_1) + \text{Var}(b_2) + \text{Var}(b_3) \\ &\quad - 2\text{cov}(b_1, b_2) - 2\text{cov}(b_2, b_3) + 2\text{cov}(b_1, b_3) \\ &= 3 + 4 + 3 - 2(-2) - 2(0) + 2(1) \\ &= 16 \end{aligned}$$

$$\therefore |t| < 2$$

$\therefore$ Do not reject H_0